

1. How can you determine the number of items in a shopping cart represented as a list in Python? (2)

2. What is the difference between "del" and "remove" in Python lists? (2)

3. In a student attendance system, a teacher is checking the attendance of students one by one from a list.

- If the teacher finds a student named "John," they stop checking further and mark attendance as complete.
- If a student's name starts with "A," the teacher skips them and moves on to the next student.

Generate python program for the above problem statement (4)

4. What is the difference between for loop and a while loop in python? Give an example of each? (3)

5. Write a Program to extract each digit from an integer in the reverse order. For example, If the given int is 7536, the output shall be "6 3 5 7", with a space separating the digits. (4)

6. You are managing an employee database with a table `Employees` containing columns `EmployeeID`, `Name`, `Department`, `Salary`, and `JoiningDate`. Write a query to retrieve the names of employees from the `Sales` department with a salary greater than ₹60,000. (5)

7. An electricity company charges its customers based on their monthly electricity usage. The charges are as follows:

- For usage up to 100 units: ₹5 per unit
- For usage between 101 to 300 units: ₹7 per unit
- For usage above 300 units: ₹10 per unit

Write a Python function that accepts a dictionary containing customer names and their electricity usage, then returns a dictionary with their calculated bill amounts. (5)

8. A company wants to calculate the bonus for its employees. The bonus is 10% of the salary for employees earning less than ₹50,000 and 5% for others. Write a Python function that takes a dictionary of employee names and their salaries and returns a dictionary with the calculated bonuses. (5)

9. Write a program to find the second largest number in a list. (3)

10. What is the output of `print list1 + list2`, if `list1 = [ 'abcd', 786 , 2.23, 'john', 70.2 ]` and `list2 = [123, 'john']`? (2)

11. Create a Python program that accepts a password from the user and raises an exception if the password is shorter than 8 characters. (5)

12. You are handling the database of a bookstore. The table `Books` contains `BookID`, `Title`, `Author`, `Genre`, and `Price`.

Write a query to find the titles of books belonging to the 'Science Fiction' genre priced below ₹500. (5)

13. In a table Employees (EmployeeID, Name, Department, Salary, JoiningDate), write a query to find the **names of the top 3 highest-paid employees** in the 'Marketing' department. (5)

14. In a table Accounts (AccountNumber, AccountHolder, Balance, AccountType), write a query to list all AccountHolders whose balance is **more than the average balance** of all accounts. (5)

15. You are working with an Employees table: EmployeeID, Name, Department, Salary, JoiningDate (5)

- Retrieve the names of all employees working in the **'Sales'** department.
- Find the total number of employees who earn more than ₹60,000.
- Display the names and salaries of employees who joined **before 2020**.
- Retrieve the **highest salary** in the company.
- Display the number of employees in each department.

16. Given a table Products with the columns: ProductID, ProductName, Category, Price, Stock, (5)
AddedDate.

- Retrieve all products that have a **stock of less than 10**.
- Display the **total number of products** in the 'Electronics' category.
- Retrieve the name of the **most expensive product**.
- Calculate the **total inventory value** (Price × Stock).

17. Design an employee management system with multiple related classes. (5)

- Create a parent class Employee with attributes name and salary.
- Add a method to **display details**.
- Create subclasses Manager (attribute: department) and Developer (attribute: programming language).
- Override the display method in both subclasses.
- Create and display objects for both subclasses.

18. Develop a result management system for a school. (5)

- Create a base class Student with attributes: name, roll_no.
- Add a method to **display basic student details**.
- Create a subclass Result with marks in three subjects.
- Write a method to calculate total marks and percentage.

19. You have a Sales table with columns: SaleID, ProductName, SaleAmount, SaleDate, Region.

Write SQL queries to (5)

- Retrieve all sales where the **SaleAmount exceeds ₹10,000**.
- Find the **total sales amount for each region**.
- Identify the **product(s) with the minimum SaleAmount**.
- Count how many sales were made in the year **2023**.

20. A weather station records temperatures in Celsius. Write a Python program that accepts a list of temperatures in Celsius and converts them to Fahrenheit using the formula $F = (C * 9/5) + 32$. Print both the Celsius and Fahrenheit values. (5)

